

Future of solar energy potential in a changing climate across the world: A CMIP6 multi-model ensemble analysis

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ARTICLE INFO

Article history:

Received 16 January 2022

Accepted 6 February 2022

Available online 23 February 2022

Keywords:

Climate change

Renewable energy

Sustainable development goals (SDGs)

Photovoltaic (PV) potential

Concentrated solar power (CSP)

CMIP6

ABSTRACT

Sustainable Development Goals (SDGs) of the United Nations (UN) underline the importance of harnessing renewable energy sources for enhancing access to clean energy without compromising the global emission goals. This study explores the impact of climate change on global solar energy potential in the near- (2015–2040) and far-future (2041–2100). Simulated energy variables from five General Circulation Models (GCMs) participating in the Coupled Model Intercomparison Project phase 6 (CMIP6) for three different Shared Socioeconomic Pathways (SSPs) – SSP1-2.6, SSP2-4.5 and SSP5-8.5 are used for the assessment. It is found that there is a 6–10% decrease (with respect to 1981–2014 climatology) in Photovoltaic (PV) potential in the Indian subcontinent and China in the boreal autumn that is possibly linked to increased post-monsoon cloud cover. A consistent decrease is noticed in North America and Australia, whereas in Europe, the projected decrease in PV potential, even for the worst emission scenario (SSP5-8.5), is restricted only to the boreal winter season, thereby posing no real threat to future PV power planning. However, a mild decrease in PV potential in Africa during austral summer and a consistent decrease in Concentrated Solar Power (CSP) all over the world contradict earlier studies.

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1. Introduction

The combustion of fossil fuels is one of the largest producers of greenhouse gases that directly contributes towards the global warming and anthropogenic climate change [1–3]. The Sustainable Development Goals (SDGs) of the United Nations (UN) emphasize on the global access to clean energy (SDG-7) and the action to combat climate change (SDG-13), and therefore, provide a strong impetus to enhance international cooperation in harvesting renewable energy sources [4]. Apart from the high greenhouse gas emissions, fossil fuel based thermal power generation systems have other limitations in terms of transmission and distribution [1]. Several developing countries still have very low electricity coverage and distribution; for example, about 630 million people in Africa and around 244 million people in India do not have access to electricity [1,5]. In vast countries with an underdeveloped hinterland, the expansion of the thermal power grid to remote areas is practically challenging. On the other hand, solar power can be generated by decentralized installation of small roof top solar

panels for domestic and community utilization [1]. Moreover, the solar power can also be harvested through large scale technologies using Concentrated Solar Power (CSP) plants, which use high temperature phase change thermal storage systems [6,7]. The abundance of sunny days (~300 days per year) in most tropical countries in Africa and Asia is another motivation for harnessing the potential of solar energy [8]. Thus, solar energy has the potential to address several of the concerns faced by the global community in its attempt in harnessing a clean, economic energy source for the rising energy demand and thereby decoupling the energy sector emissions from economic growth [9]. However, before a large scale investment in solar energy development systems through such photovoltaic (PV) modules or CSP plants, the policy makers need to gauge the sensitivity of solar energy potential considering the projected future climate change [10], which is a major concern in the current age.

The amount of solar radiation, which is influenced by the absorption and scattering via clouds and aerosol concentration, is a direct measure of PV potential [11]. Factors like air temperature,

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relative humidity and wind speed also influence the PV potential [12–15]. All these meteorological factors are subjected to fluctuations with alteration in the frequency of weather and climate extremes, of warm and cloudy days etc. [2]. In fact, if the number of days per year with very low PV output increases due to the change in solar radiation, then there would be an imbalance in the energy production and consumption, leading to a greater expenditure on the energy storage and grid stabilization [10]. Hence, for the economic sustainability of PV/CSP projects, the enhanced climate and weather variability associated with different future climate change scenarios needs to be considered.

Globally, an overall decrease in the mean PV power during 2006–2100 is projected, except an increase in East Asia, Europe, Central Africa and Central America [16]. However, there is a substantial spatial variation in the projected changes even within a given region. For example, southern and western Europe is expected to experience an increase in PV potential, whereas northern and eastern Europe is expected to experience a decrease during the middle to end of 21st century under different Intergovernmental Panel on Climate Change – Special Report on Emissions Scenarios (IPCC SRES) [17,18]. A decrease in PV potential and solar radiation is also predicted in the West Africa over the 21st century under varied Representative Concentration Pathways (RCPs) [2,19–22]. At the seasonal scale, most South African countries are projected to experience a decrease in solar radiation during June through August, but an increase of the same during December through February [23,24]. Increase in PV output is mostly predicted for the regions with expected decrease in cloud cover and increase in clear sky radiation, such as south-eastern parts of North America, whole of China and parts of Europe [25]. Studies indicated a decrease in PV potential in western US and Saudi Arabia and an increase in China between 2010 and 2080 [26]. Little to no change in PV output was projected across Australia during the same period; however, a marginal increase in the percentage of the CSP output was projected. It was also noted that the changes in CSP were more sensitive to climate change than PV and almost entirely caused by the corresponding changes in solar radiation [26].

However, there is a lack of concurrence among the findings related to changing solar energy potential due to the difference in the climate models considered and the scale of analysis. Moreover, several studies reported the results from a single Global Climate Model (GCM) or Regional Climate Model (RCM) that definitely limits their reliability. Most of the findings from the early years of the last decade are based on coarse resolution GCM simulations under SRES A1B or B2 scenarios [18,25,26] or derived from GCMs, participating in the Coupled Model Intercomparison Project phase 5 (CMIP5) consortium, under different RCPs [2,10,22]. However, availability of the high-resolution simulations and improved Shared Socio-economic Pathways (SSPs), from the latest phase (phase 6) of CMIP (i.e., CMIP6), necessitates the re-assessment of the sensitivity of PV potential and CSP output in a changing climate. Moreover, a global analysis is also required for a holistic view and comparative conclusions between continents. It is imperative to perform a multi-model ensemble analysis based on the latest climate projections available to the research community to study the spatial and temporal variability of the relevant energy variables over the 21st century. Hence, this study examines the impact of climate change on global solar energy potential in the near-future (2015–2040) and far-future (2041–2100) with respect to the historical period (1981–2014). The percentage changes in PV potential and CSP, calculated using key variables for renewable energy, simulated by five different GCMs, participating in the CMIP6 for three different SSPs, i.e., SSP1-2.6, SSP2-4.5 and SSP5-8.5 are evaluated for four seasons – boreal spring, summer, autumn and winter, also designated as seasons A, B, C and D, respectively.

2. Data

The monthly estimates of Surface Downwelling Shortwave (SW) Radiation (I) (W/m^2), near-surface (at 2 m height) Air Temperature (T) ($^{\circ}\text{C}$) and near-surface (at 10 m height) Wind Speed (WS) (m/s) from five different GCM outputs, participating in CMIP6, are used. The CMIP6 is an expansion over CMIP5 as there are greater number of modelling groups and also greater number of future scenarios examined. The CMIP6 climate models are driven by a new set of emissions scenarios based on different socioeconomic assumptions, designated by different SSPs. The RCPs, which designate different possible future greenhouse gas concentration by the IPCC's 5th Assessment Report (AR5), are updated as new versions in the SSPs, such as SSP1-2.6, SSP2-4.5, SSP4-6.0, and SSP5-8.5. In this study, SSP1-2.6, SSP2-4.5 and SSP5-8.5 are used for representing the 'best', 'moderate' and 'worst' scenarios of radiative forcing in 2100. The five GCMs are selected based on the availability of the relevant energy variables during the historical (1981–2014) as well as future period (2015–2100) for all the selected SSPs. Table 1 presents a summary of the GCMs along with their spatio-temporal resolution. As the different model outputs are available at different spatial resolutions, they are all regridded to a common resolution of $0.25^{\circ} \times 0.25^{\circ}$ (latitude $\times$ longitude) using bilinear interpolation method [27] as used in many studies [28–30]. Apart from the three aforementioned variables which are directly used to compute PV power output, the Cloud Area Fraction (f_{clt}) is also obtained from each GCM. The ' f_{clt} ' denotes the percentage of cloud (both large-scale and convective) in the whole atmospheric column, as seen from the earth surface (upward) or from the top of the atmosphere (downward). Hence, it also influences the solar radiation, and in turn, the PV potential.

In order to assess the reliability of the CMIP6 GCM datasets, the ERA-5 reanalysis monthly averaged gridded (0.25° lat $\times$ 0.25° lon) data of Air Temperature (T), Solar Radiation (I) and near-surface Wind Speed (WS) developed by the European Centre for Medium Weather Forecasts (ECMWF) has been used (<https://www.ecmwf.int/en/forecasts/datasets/reanalysis-datasets/era5> accessed in May 2021).

3. Methodology

3.1. Computation of PV potential

The PV potential represents the fraction of the power output under standard conditions that a PV module may exhibit in the field [10,17]. PV may be calculated by power rating method or energy rating method. In the power rating method, the instantaneous photovoltaic power generation is integrated over time, whereas, in the energy rating method, photovoltaic potential is estimated by multiplying the total solar irradiance during a specific period of time by a performance ratio [31]. The later method is adopted in most recent studies [2,10,17,21,22] and is also adopted in the present study. The monthly estimates of PV potential are computed from the monthly estimates of the variables I , T and WS, obtained from 5 GCMs for each of the 3 SSPs. For a given GCM and SSP combination, the monthly PV potential is given by

$$PV_{pot} = P_R \frac{I}{I_{STC}} \quad (1)$$

where I is the SW radiation striking on PV modules in field conditions, I_{STC} is the SW radiation applied to the module under standard test conditions (STC), which is 1000 W/m^2 , and P_R is the performance ratio, that accounts for the effect of the cell temperature (T_{cell}) [10,22]. P_R is given by

Table 1
Summary of the five GCMs from CMIP6.

Model Name	Model Centre	Grid Size (lon × lat) (Spatial Resolution)			Temporal Resolution
		Historical	Future	Re-gridded	
HadGEM3-GC31	National Centre for Meteorological Research, France	1024 × 768 (~ 0.35° × 0.23°)	432 × 324 (~ 0.83° × 0.55°)	1440 × 721 0.25° × 0.25°	Monthly
EC-Earth3	EC-Earth Consortium, Europe	1024 × 512 (~ 0.35° × 0.35°)	512 × 256 (~ 0.70° × 0.70°)		
IITM-ESM	Centre for Climate Change Research, Indian Institute of Tropical Meteorology, India	192 × 94 (~ 1.87° × 1.91°)	192 × 94 (~ 1.87° × 1.91°)		
GFDL-ESM4	National Oceanic and Atmospheric Administration, Geophysical Fluid Dynamics Laboratory, USA	288 × 180 (~ 1.25° × 1.00°)	288 × 180 (~ 1.25° × 1.00°)		
CanESM5	Canadian Centre for Climate Modelling and Analysis, Canada	128 × 64 (~ 2.81° × 2.81°)	128 × 64 (~ 2.81° × 2.81°)		

$$P_R = 1 - \gamma(T_{cell} - T_{STC}) \quad (2)$$

where T_{cell} is the cell temperature at the field conditions, T_{STC} is the cell temperature under STC, which is 25 °C, and γ depends on the type of solar cell. γ may be conveniently taken as 0.005°C⁻¹ for monocrystalline silicon solar cell, which is a popular used cell type [10,22,32]. T_{cell} may depend on solar radiation, air temperature, wind speed and relative humidity [33]. At a regional scale, some of these variables may have a relatively smaller contribution to the changes in PV potential [2,21]. For the global analysis in this study, T_{cell} is expressed as a function of the three variables I , T and WS , as provided in eqn (3). While I and T increase the cell temperature, WS lowers it due to its cooling effect.

$$T_{cell} = c_1 + c_2 T + c_3 I - c_4 WS \quad (3)$$

where $c_1 = 4.3$ °C, $c_2 = 0.943$, $c_3 = 0.028$ °C W⁻¹ m² and $c_4 = 1.528$ °C m⁻¹ s are used for monocrystalline silicon solar cell, based on previous studies [2,26].

3.2. Computation of Concentrated Solar Power (CSP)

In large scale solar energy systems, electricity is produced by concentrating the direct solar irradiance through a combination of lenses. CSP is estimated as

$$CSP = \eta_{CSP} I_d \quad (4)$$

where η_{CSP} is the thermal efficiency of the CSP and I_d is the direct solar irradiance [26]. As I_d is not generally available from the model outputs, I_d may be computed using the empirical equation [34].

$$I_d = 0.75 I_{sc}(1 - f_{clt}) \quad (5)$$

where I_{sc} is the clear sky irradiance at the surface and f_{clt} is the fractional cloud cover and the coefficient 0.75 represents the effect of scattering of sunrays by air molecules and aerosols, which occur even under clear sky conditions. As I_{sc} is also not available from the model outputs, it is obtained using the following expression [35].

$$f_{clt} = 1 - I/I_{sc} \quad (6)$$

where I is the solar irradiance. Thus,

$$I_d = 0.75 I \quad (7)$$

The thermal efficiency η_{CSP} in eqn (4) is given by

$$\eta_{CSP} = k_0 - k_1 \frac{(T_i - T)}{R_d} \quad (8)$$

where $k_0 = 0.762$ W m⁻² °C⁻¹ and $k_1 = 0.2125$ W m⁻² °C⁻¹ are the coefficients specific to the solar energy collector [2,26,36]. Further, $T_i = 115$ °C is the temperature of the heat transfer fluid (molten salts) and T is the ambient temperature.

In the present study, all the equations in the preceding two subsections have been borrowed from the latest available literature. However, the detailed quantitative relationship of PV and CSP with cloud cover may be explored further for a better assessment. This may be considered as a potential future scope of study in this research domain.

3.3. Seasonal analysis

As the number of daylight hours (which is season-dependent) is related to the PV potential, it is sensible to perform the analysis season-wise. However, the boreal seasons are opposite to the austral seasons. Moreover, the tropical regions have a distinct monsoon season, which is not applicable for the temperate regions. Hence, for this global study, the four seasons i.e., boreal spring (Mar-Apr-May, MAM), boreal summer (Jun-Jul-Aug, JJA), boreal autumn (Sep-Oct-Nov, SON) and boreal winter (Dec-Jan-Feb, DJF) are considered. These seasons are denoted as season A, B, C and D, respectively. The changes in the PV potential and CSP are examined for each of the aforementioned seasons from five GCMs under CMIP6 for 3 different SSPs.

4. Results

4.1. Agreement of CMIP6 model simulations with respect to ERA5 reanalysis products

We gauged the efficiency of the CMIP6 GCMs (details available in the Data section) to produce reliable estimates of the three key variables for renewable energy, i.e., temperature (T), downwelling shortwave radiation (I) and wind speed (WS) through a comparison with ERA5 reanalysis products (Fig. 1). The general agreement of the values with low bias (within ±5% for T and WS) provides sufficient reliance on the future projections of the solar energy

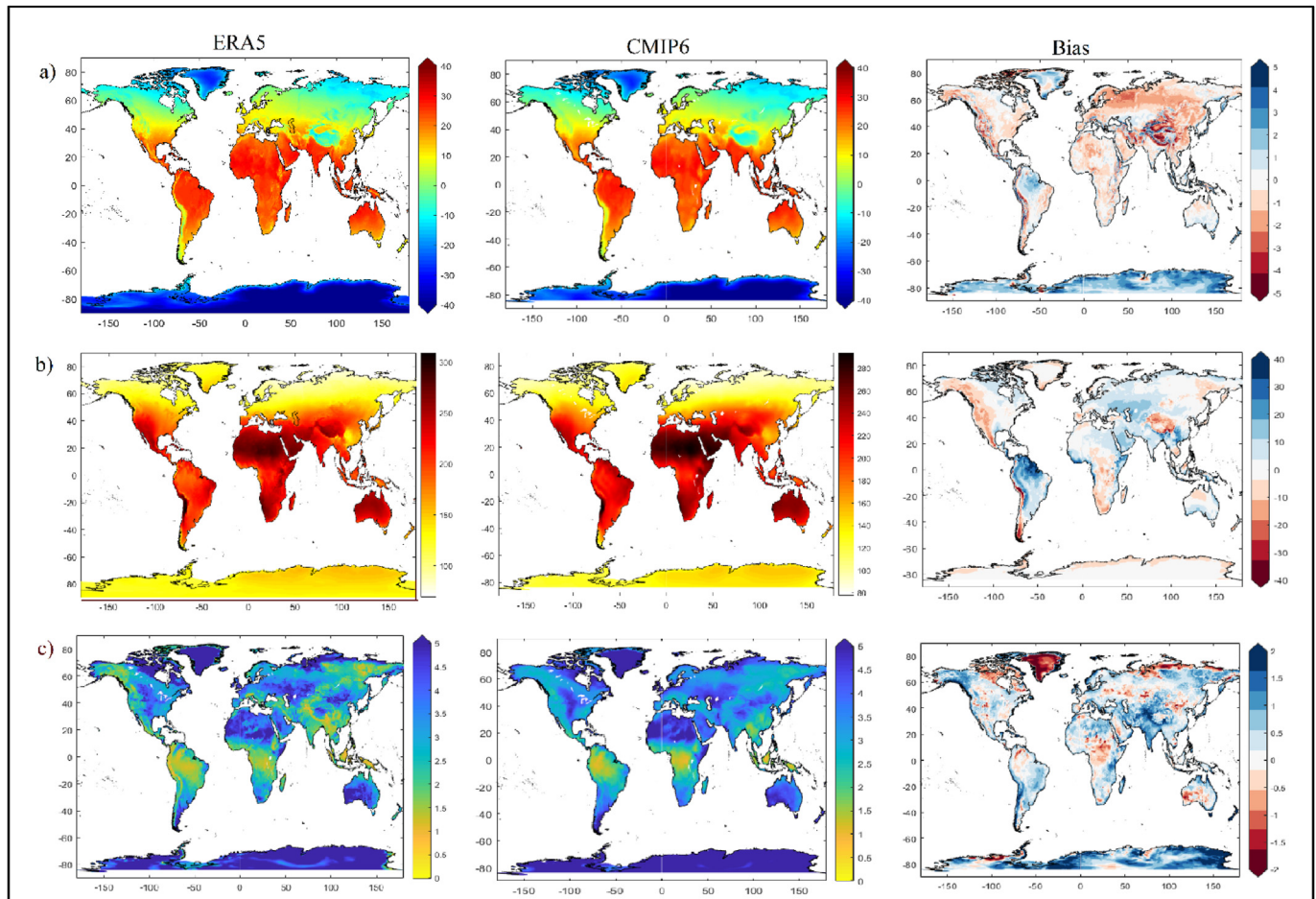


Fig. 1. Correspondence between CMIP6 simulations and ERA5 reanalysis products. Across the globe, the CMIP6 model averaged values are in agreement with the ERA5 reanalysis variables used to calculate the PV potential and CSP during the historical period of 1981–2014. Reliable estimates of the energy variables namely, a) temperature (in °C denoted by T), b) downwelling shortwave radiation (in W/m² denoted by I) and c) wind speed (in m/s denoted by WS) provides more confidence in the future projections of the solar energy resources. Both T and I show negative bias in Central Asia and positive bias in northern parts of South America whereas WS shows a positive bias in the Himalayan region in Asia.

resources. Although the bias in I is much higher (>30%) in Venezuela, Guyana and northern Brazil, it is moderate (within $\pm 10\%$) everywhere else.

4.2. Change in PV potential in the near- and far-future

Globally, the percentage changes in PV potential during the near-future period (2015–2040) with respect to the historical period (1981–2014) are more prominent for the boreal autumn and winter as compared to the boreal spring and summer for all three SSPs (Fig. 2). Apart from the regions beyond the Arctic Circle (where simulations have limited reliability), the largest changes are observed in the Indian subcontinent and China in boreal autumn, with a decrease of about 5% (under SSP1-2.6) to 10% (under SSP5-8.5). This may be directly attributed to the decrease in solar radiation (varying between 1% and 7%) in that region (Supplementary Figure 1). This may be also partially supported (at least for SSP1-2.6 and SSP5-8.5) by the increasing cloud cover percentage (f_{clt}) (Supplementary Figure 2) over the Himalayan belt, northern India and China during boreal autumn, which coincides with the post-monsoon season in India. A decrease (2–4%) in PV potential is evident throughout the Middle East and Asia in boreal autumn under SSP2-4.5 and SSP5-8.5 (Fig. 2); however, the spatial extent is much less for SSP1-2.6 and for all SSPs in boreal winter. Similar

observations hold in the African countries (Niger, Chad, Sudan, Ethiopia, South Sudan, Central African Republic, Kenya, Angola, Zambia, Namibia) during the same seasons and SSPs. The decrease in PV in the aforementioned regions may be considered to be a reflection of the corresponding decrease in solar radiation in the boreal autumn and winter (Supplementary Figure 1). The direction of observed changes in PV potential in India and Asia is not as well supported by the increasing air temperatures as by the decreasing wind speed that decreases the cooling effect (Supplementary Figure 3 and 4). The relationship with cloud cover is not very direct since there is an increase of 6–8% (decrease of 4–10%) in f_{clt} in the eastern (western) parts of Africa (Supplementary Figure 2). During boreal spring, summer and even autumn, a clear increase in PV potential is observed across Europe for all the SSPs, with gradual increase in areal extent from SSP1-2.6 to SSP2-4.5 to SSP5-8.5 (Fig. 2).

However, in boreal winter, PV potential is found to decrease across Europe. The increase (decrease) in PV potential in boreal summer and autumn (boreal winter) is concurrent with a simultaneous decrease (increase) in f_{clt} (Supplementary Figure 2), but the same is not evident in boreal spring. The increase in boreal spring is supported by increased radiation (Supplementary Figure 1) that may be the result of a decrease in RH, as drier spring and summer are predicted in Europe, particularly for higher emission scenarios [37].

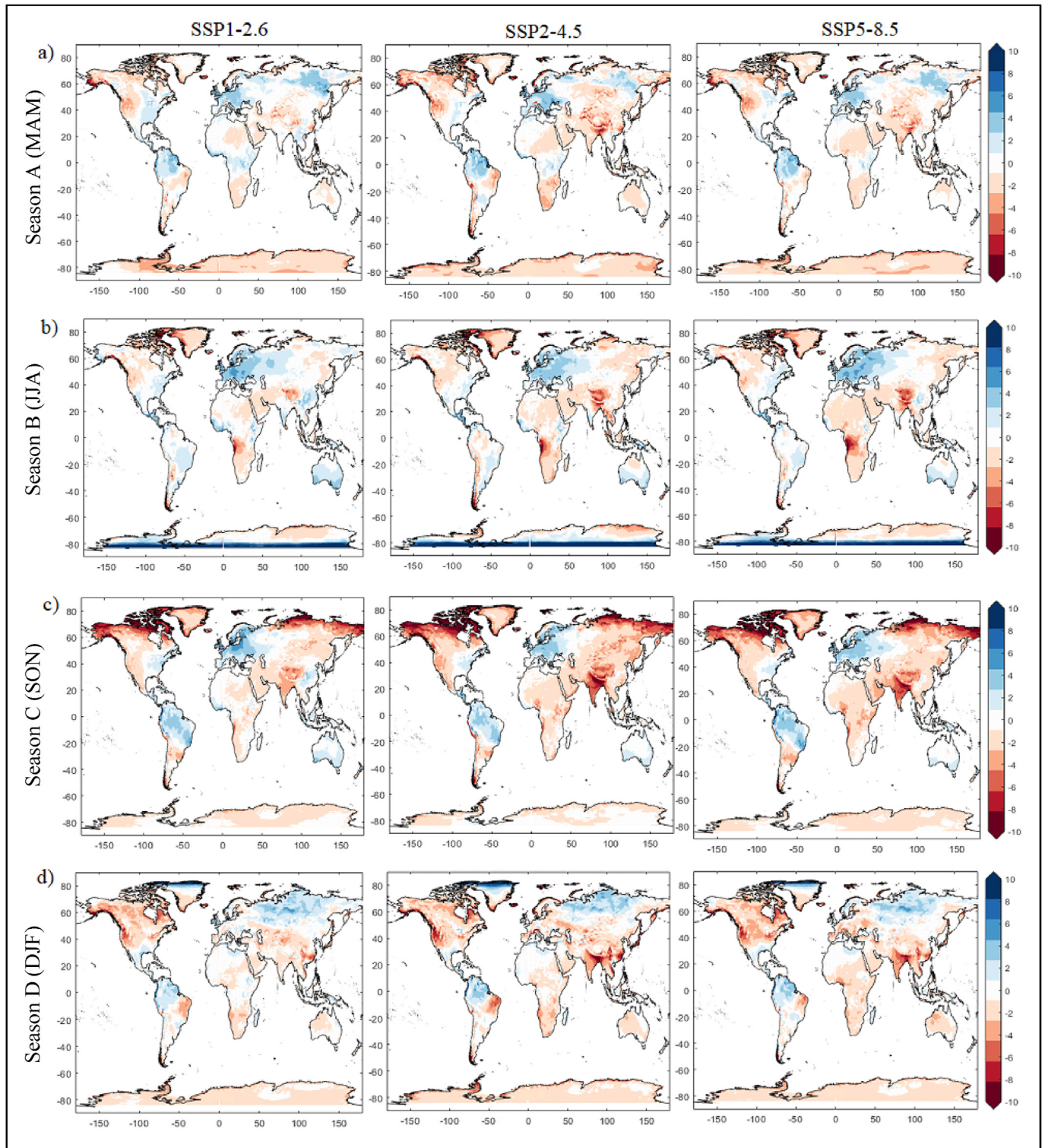


Fig. 2. Model averaged change in PV potential during the near-future using CMIP6 model projections. Percentage change during the period of 2015–2040 (near-future) with respect to the period of 1981–2014 (historical) under 3 different SSPs – SSP1-2.6 (lowest emission scenario), SSP2-4.5 (moderate emission scenario) and SSP5-8.5 (highest emission scenario) for, a) boreal spring (Season A: Mar–Apr–May), b) boreal summer (Season B: Jun–Jul–Aug), c) Boreal autumn (Season C: Sep–Oct–Nov) and d) Boreal winter (Season D: Dec–Jan–Feb). The changes are more prominent for the boreal autumn and boreal winter as compared to the boreal spring and boreal summer.

The western part of the North American continent consistently shows a decrease in PV potential (maximum decrease in boreal autumn, 4–5%) for all the SSPs for all the seasons, which can be supported by the prominent increase in cloud cover

(Supplementary Figure 2), although the spatial distribution patterns match rather crudely for the boreal summer and autumn. The eastern part of the North American continent shows a marginal increase in the PV potential (~2%) which reduces in spatial extent in

boreal spring, summer and autumn and disappears in boreal winter and does not reflect the changes in cloud cover. In the South American continent, there is a clear increase (2–4%) in PV potential in all the seasons except austral winter, when there is almost no change detected. In Australia, the change is positive (2–4%) in austral autumn and summer and mild negative (within 2% and much lesser spatial extent) in austral winter and spring. In both these southern hemisphere continents, the changes in the PV potential are corroborated well in the austral autumn with a corresponding decrease in cloud cover (Supplementary Figure 2); however, the correspondence with cloud cover is missing in the other three seasons, which is also observed for the decrease in the PV potential across the African continent in austral spring and summer. Such discrepancies may be related to changes in other factors, such as RH or mineral dust aerosols; for example, dust loadings are projected to increase in Africa, particularly in summer [38,39].

During the far-future period, for boreal spring and autumn, there is a distinct increase in the PV potential across Europe, while in boreal winter, there is a sharp decrease, which is strongest for SSP5-8.5 (Fig. 3). In boreal spring, the three SSPs indicate a disagreement in the directions of change in PV potential – increase (SSP1-2.6), ‘slight increase to no change’ (SSP2-4.5) and ‘no change to decrease’ (SSP5-8.5). Such disagreements across the different SSPs are also evident in India and Africa for all the seasons. The seasonal increase (decrease) in the PV potential (Fig. 3) are also corroborated by a corresponding increase (decrease) in solar radiation (Supplementary Figure 5) and decrease (increase) in cloud cover (Supplementary Figure 6). North America (particularly the north-western part) shows a consistent decrease in the PV potential for all the seasons, with the largest decrease in boreal winter and the least decrease in boreal summer. Similarly, South America shows an increase in PV potential, which is prominent in austral spring and summer, but weak in austral winter. Australia indicates decrease in PV potential for all cases except in austral winter. The spatial patterns showing the increase (decrease) are explained by the corresponding increase (decrease) in solar radiation (Supplementary Figure 5) and decrease (increase) in cloud cover (Supplementary Figure 6) for the Americas, but not so much in case of Australia. Air temperature has limited use in explaining the variability in the PV potential since it is noticed to increase globally (Supplementary Figure 7).

The most prominent decrease in the PV potential is projected during boreal winter for SSP5-8.5 with a large decrease (>10%) over Canada and US in North America and western and central Europe (Nordic countries, Germany, Poland, Belarus, western Russia) and up to 4% decrease in Asia (India, China) and Australia and up to 10% decrease in Africa (Nigeria, Cameroon, Republic of Congo and the Democratic Republic of Congo). The only increase in the PV potential for this combination (season and SSP) is observed in the South America (Colombia, Venezuela, Guyana and western Brazil).

4.3. Change in CSP in the near- and far-future

Compared to the PV potential, the CSP projections indicate much less spatial and seasonal variability and are projected to decrease throughout the world for both near-future (Fig. 4) and far-future (Fig. 5) for all the seasons. In the near-future, CSP shows a decrease between 1 and 2% everywhere except in Russia (eastern Eurasia) (which shows no change) for the boreal spring and summer for all the SSPs (Fig. 4). A decrease in CSP of up to 4% is observed in Colombia, Venezuela, Guyana and Suriname of South America (boreal spring) and a few regions in Sudan in Africa (boreal summer). Both South America (Brazil and Bolivia) and North America (Canada) show pronounced decrease in CSP in boreal autumn and

the later also in boreal winter. Eastern Eurasia continues to show little to no-change in CSP in boreal autumn and winter, respectively. For far-future (Fig. 5), the magnitude of decrease is much higher (4–6%) as compared to that in the near-future, particularly in case of the worst emission scenario, i.e., SSP5-8.5. For all the seasons, the least decrease is projected for SSP1-2.6 (most sustainable pathway with lowest emission scenario) and the largest decrease (>6%) is projected in SSP5-8.5 (worst scenario). The strongest decrease is noticed in Brazil and Venezuela (boreal autumn, SSP5-8.5) and in the Middle East and North African (MENA) countries (boreal summer, SSP5-8.5).

5. Discussion

The projected changes in solar radiation support the general increase in PV potential in the near-future period across Europe, South America, Australia and eastern parts of North America, as well as the decrease across several regions of Asia, Africa and western parts of North America. The changes in the PV potential projected towards the end of the century during boreal summer in Europe, Asia and Australia for SSP2-4.5 reaffirm the findings from a previous season-wise analysis using simulations from CMIP5 models; however, the mild decrease in Africa during austral summer contradicts earlier findings [10].

Furthermore, the changes in the PV potential are consistent with the findings from other studies for boreal winter for the Americas and Australia [10]. However, there is a notable difference for western Europe.

In Europe, the projected far-future decrease in PV potential, even for the worst emission scenario (SSP5-8.5), is predicted only for the boreal winter season and for the Nordic countries. In these particular cases, solar energy is not expected to be the chief contributor to the development of renewable energy and hence do not threaten the low-carbon economy planning [40]. This finding is in general agreement with previous studies using RCP8.5 simulations [17]; although a direct comparison in the percentage of change is not suitable due to the differences in scale of analysis, climate change scenarios and base climatology adopted in the available studies.

In the African continent, the decrease in mean PV potential is projected to be the highest (more than 10%) in austral winter in the countries of Nigeria, Republic of the Congo, Angola during the near-future and extend over a larger part of the peninsula towards the end of the century under the worst scenario (SSP5-8.5). Previous findings report a much smaller decrease (in the order of 2%) in the mean PV values [21,22]. However, the present findings during austral winter may be more appropriately compared with the lowest (rather than mean) PV values, i.e., 5th quantile, which were earlier found to decrease by 8% [21].

Predicted changes in PV potential in India, China and the rest of Asia are similar in direction with previous studies, however, the magnitude of decrease is much smaller than that predicted earlier [26]. Only the far-future projections for the worst climate change scenario (SSP5-8.5) indicate a larger percentage of decrease and reflect the figures reported in earlier studies even under moderate scenarios [26]. Thus, only under the SSP5-8.5 scenario, the altered PV potential could be a threat to PV power planning in Southern Asia.

In case of CSP outputs, the changes are negative almost everywhere for all the climate scenarios (SSPs). This is in contrast to previous works indicating increase in CSP in China, Europe and Australia under SRES A1B scenario [26]. Moreover, direct correspondence of CSP output changes with solar radiation is not explicit, unlike in previous studies [26].

Here, we use the simulation outputs from 5 GCMs, participating

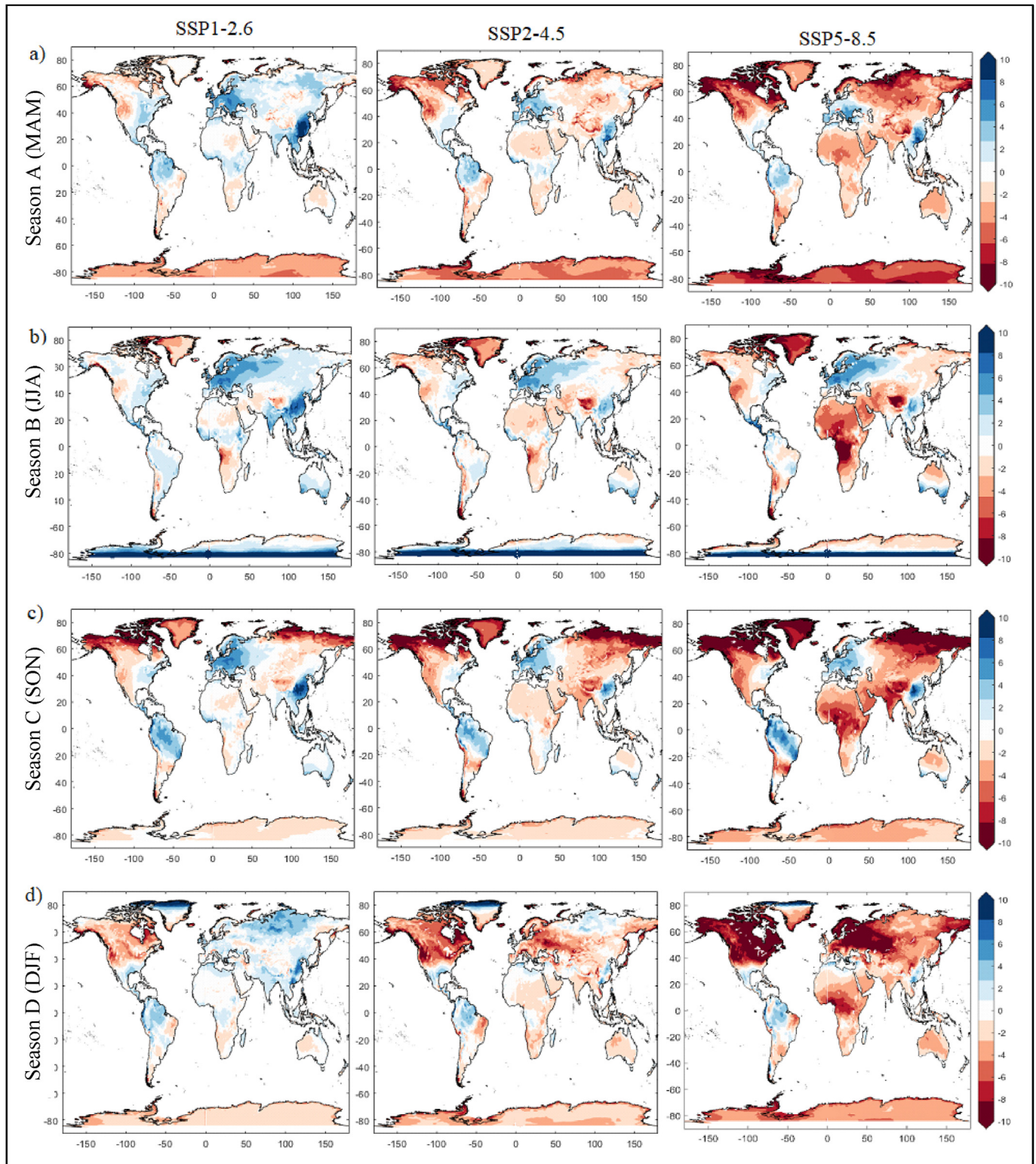


Fig. 3. Model averaged change in PV potential during the far-future using CMIP6 model projections. Percentage change during the period of 2041–2100 (far-future) with respect to the period of 1981–2014 (historical) under 3 different SSPs – SSP1-2.6, SSP2-4.5 and SSP5-8.5 for, a) boreal spring (Season A: Mar–Apr–May), b) boreal summer (Season B: Jun–Jul–Aug), c) boreal autumn (Season C: Sep–Oct–Nov) and d) boreal winter (Season D: Dec–Jan–Feb). The changes are more prominent for the boreal autumn and boreal winter as compared to the boreal spring and Boreal summer under SSP1-2.6 and SSP2-4.5. The most intense decrease in the PV potential is projected in boreal winter for SSP5-8.5 with a large decrease (>10%) over the whole of North America and western Europe and up to 4–10% decrease in Asia, Africa and Australia.

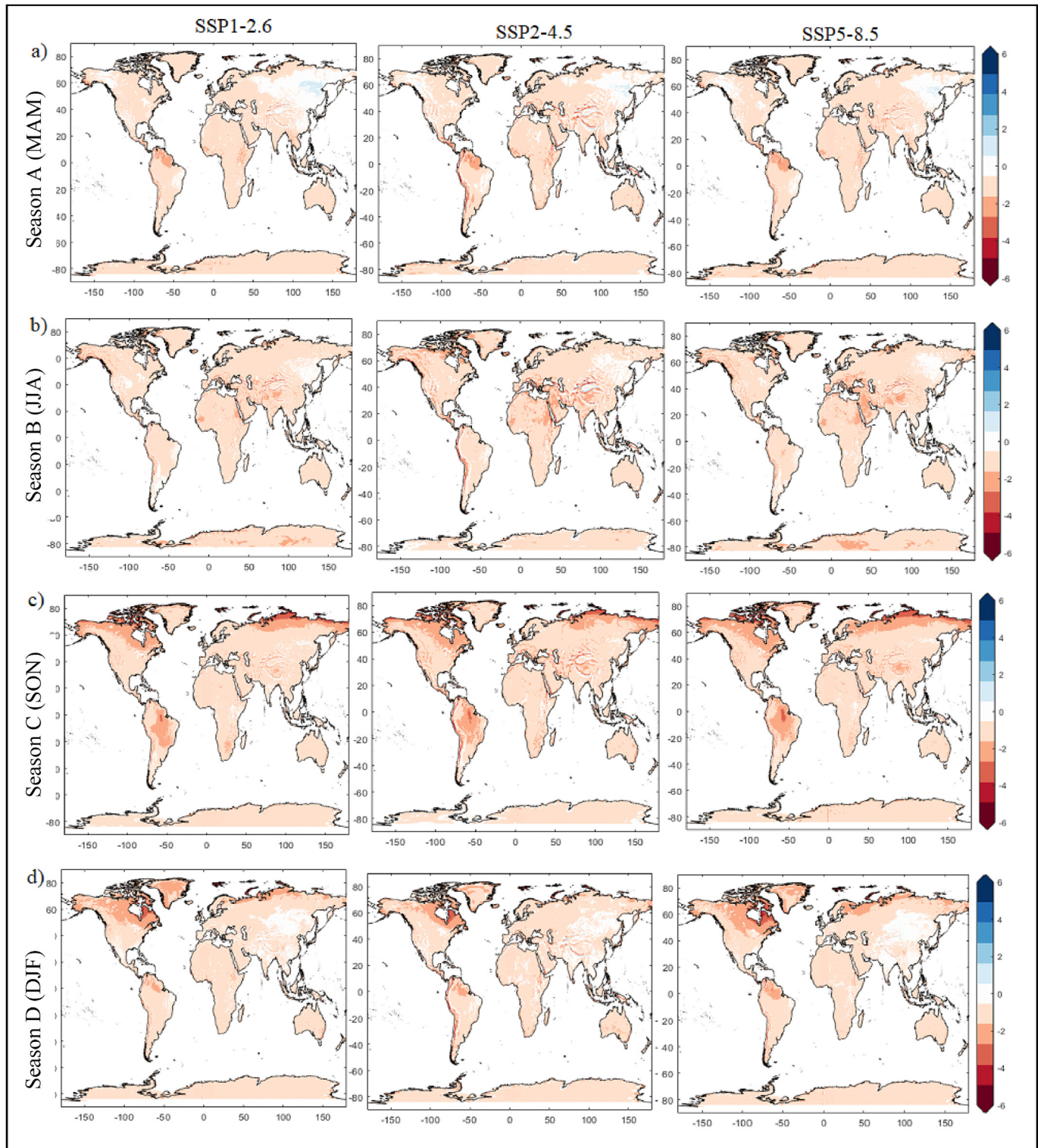


Fig. 4. Model averaged change in CSP during the near-future using CMIP6 model projections. Percentage change during the period of 2015–2040 (near-future) with respect to the period of 1981–2014 (historical) under 3 different SSPs — SSP1-2.6, SSP2-4.5 and SSP5-8.5 for, a) boreal spring (Season A: Mar–Apr–May), b) boreal summer (Season B: Jun–Jul–Aug), c) boreal autumn (Season C: Sep–Oct–Nov) and d) boreal winter (Season D: Dec–Jan–Feb). The CSP shows a decrease between 1 and 2% throughout the globe except in the eastern part of Eurasia (which shows no change) for boreal spring and boreal summer.

in the CMIP6 for an overall large-scale assessment of PV/CSP power planning for different continents and countries in the context of a changing climate. As more model simulations of the relevant energy variables are added to the consortium in future, they may be

incorporated to reduce the uncertainty in the ensemble model mean. Limitations of this study cannot be denied in the context of several micro-climatic factors, such as dust, convection, hailstorm, rainfall etc., that can alter the energy outputs if the operating

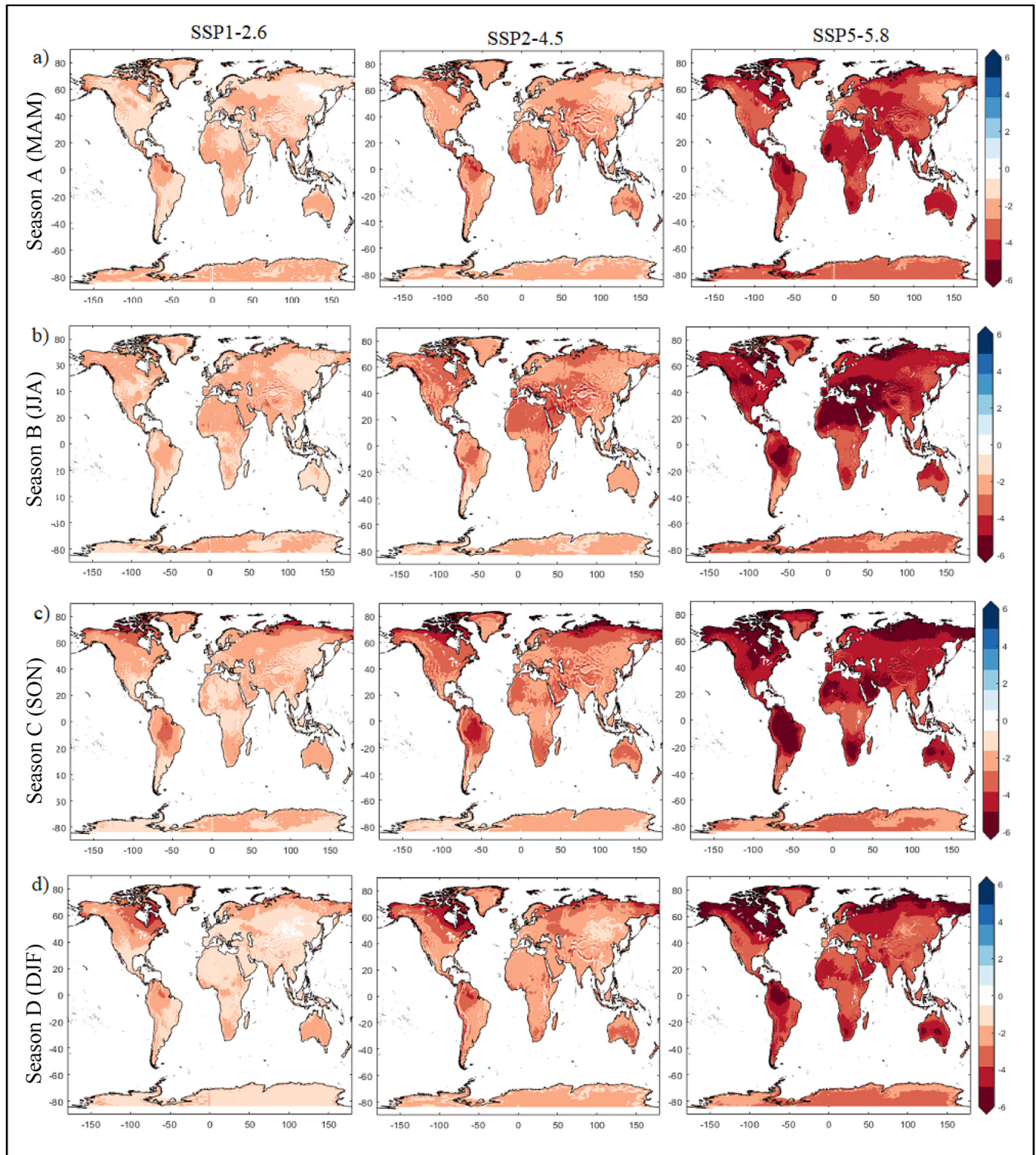


Fig. 5. Model averaged change in CSP during the far-future using CMIP6 model projections. Percentage change during the period of 2041–2100 (far-future) with respect to the period of 1981–2014 (historical) under 3 different SSPs – SSP1-2.6, SSP2-4.5 and SSP5-8.5 for, a) boreal spring (Season A: Mar–Apr–May), b) boreal summer (Season B: Jun–Jul–Aug), c) boreal autumn (Season C: Sep–Oct–Nov) and d) boreal winter (Season D: Dec–Jan–Feb). The percentage decrease is generally in the order of 2–6% across the globe with the least decrease under SSP1-2.6 (least emission scenario) and the largest decrease (>6%) under SSP5-8.5 (worst emission scenario).

conditions differ significantly from the rated conditions. Besides the wind speed, even the direction of wind is also known to affect the performance of PV plants [41]. Apart from the environmental

conditions, the energy output also depends on the type of PV technology adopted, such as crystalline silicon, amorphous silicon, dye sensitized etc [42]. Moreover, this study evaluates the changes

in PV potential and CSP outputs due to climate projections under three different future scenarios, designated by different SSPs; however, the actual path adopted by the global community in terms of economic growth and development goals could be different, particularly due of unexpected global scale events such as the current on-going pandemic.

6. Conclusions

In accordance with the SDGs (#7 and #13) of the UN, this study explores the impact of climate change on global solar energy potential in the near (2015–2040) and far-future (2041–2100) with respect to the historical period (1981–2014). The percentage changes in PV potential and CSP, calculated using simulated energy variables by five different GCMs, participating in the CMIP6 for three different climate change scenarios, i.e., SSP1-2.6, SSP2-4.5 and SSP5-8.5, are evaluated for four seasons – boreal Spring, Summer, Autumn and Winter. The major conclusions from this study are:

- 1) In the near-future (2015–2040), the changes in Photovoltaic (PV) potential varies spatially and seasonally. A general increase is observed in Europe, South America, Australia and eastern parts of North America, while a decrease is observed in several regions of Asia, Africa and western parts of North America. As air temperature is projected to increase almost everywhere, it has limited use in explaining the variability in the PV potential. However, the projected changes in solar radiation, more or less, support the direction of changes in the PV potential.
- 2) The most prominent change in multi-model mean PV potential over the near-future period (2015–2040) are observed for the boreal Autumn in the Indian subcontinent and China, with a decrease of 6–10% with respect to the historical period (1981–2014). This is partly explained by the projected increase in post-monsoon cloud cover percentage during this period.
- 3) In the near-future period, except boreal Winter, the PV potential increases (~4–6%) across all seasons with a concurrent decrease in cloud cover fraction over some parts of Europe, and a positive trend covers the whole of Europe under the SSP5-8.5 (worst emission scenario).
- 4) In the southern hemisphere continents of Australia and South America, the increase in the PV potential in the Austral Autumn are corroborated well with a corresponding decrease in cloud cover during the near-future period. However, the correspondence is missing in the other three seasons, which is also true for the decrease (2–4%) in the PV potential across the African continent in Austral Spring and Summer.
- 5) In the far-future (2041–2100), the directions of change in the PV potential are in agreement with the near-future (2015–2040) projections for the American continents. However, there is substantial variation between different SSPs for Europe, Africa, Asia and Australia, e.g., Europe in boreal spring indicates *increase/no-change/decrease* for *SSP1-2.6/SSP2-4.5/SSP5-8.5*, respectively.
- 6) Compared to the PV potential, the CSP projections indicate much less spatial and seasonal variability and are projected to decrease throughout the world for both near- and far-future periods for all the seasons. However, the magnitude of decrease is much higher for far-future (4–6%) as compared to that in near-future (2–4%), particularly in case of the worst emission scenario, i.e., SSP5-8.5.

The study is expected to be useful as a first step in planning future investment in solar energy based on the latest climate projections available to the research community. However, inclusion of more GCMs to reduce the uncertainty and investigation of micro-

climatic factors are essential to assess local scale variation in solar energy potential.

Authors' contributions

RM conceived the central idea, and KC together with RD conceptualized the formal analysis. RD, KC and RM contributed to the interpretation of results and preparation of the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

Authors wish to acknowledge the partial support from the sponsored projects supported by Department of Science and Technology and Ministry of Earth Science, Government of India (Ref No. DST/CCP/CoE/79/2017(G) and ECR/2017/001880).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.renene.2022.02.023>.

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